ETHICS OF NEW TECHNOLOGIES

W08IST-SM4017S - Seminar Assignment

Codes of Ethics, a Case Study, and a Utilitarian Analysis

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1. Codes of Ethics Selected

1.1 ACM Code of Ethics and Professional Conduct (2018)

The **Association for Computing Machinery (ACM)** published its most recent Code of Ethics in 2018, replacing the earlier 1992 version. It is the most widely recognized ethical framework for computing professionals worldwide. The code is organized into four broad categories: (1) General Ethical Principles, (2) Professional Responsibilities, (3) Leadership Principles, and (4) Compliance with the Code.

Key principles most relevant to this report include:

1.1 - Contribute to society and to human well-being, acknowledging that all people are stakeholders in computing.

1.2 - Avoid harm. Computing professionals must take care to avoid causing harm to others.

1.4 - Be fair and take action not to discriminate on the basis of, among others, race, ethnicity, or socioeconomic status.

1.6 - Respect privacy: ensure that personal data is collected with proper consent and used only for intended purposes.

2.5 - Give comprehensive and thorough evaluations of computer systems and their impacts, including unintended consequences.

2.7 - Foster public awareness and understanding of computing, its consequences, and its limitations.

1.2 IEEE Code of Ethics (2020)

The **Institute of Electrical and Electronics Engineers (IEEE) Code of Ethics** is a shorter, more concise document that sets out ten commitments for engineering and technology professionals. Unlike the ACM code, it does not focus exclusively on computing but addresses the broader field of engineering. It was most recently updated in 2020.

Key clauses most relevant to this report include:

- Clause 1 - Hold paramount the safety, health, and welfare of the public, and strive to comply with sustainable development principles.
- Clause 2 - Avoid real or perceived conflicts of interest whenever possible, and to disclose them when they do exist.
- Clause 5 - Improve the understanding of technology; its appropriate application, and potential consequences.
- Clause 7 - Seek, accept, and offer honest criticism of technical work; acknowledge and correct errors.
- Clause 8 - Treat fairly all persons and not engage in acts of discrimination based on race, religion, national origin, or other relevant characteristics.

1.3 Similarities and Differences

Both codes share a fundamental commitment to public welfare as the highest priority, rejecting the idea that professional interests or commercial pressure can override harm to the public. Both also require honesty, transparency in technical work, and respect for the dignity of all people regardless of background.

However, notable differences exist:

- Scope: The ACM code is exclusively focused on computing and software; the IEEE code covers all engineering disciplines, making it more general in language.
- Detail: The ACM code provides detailed sub-principles and extensive commentary for each principle. The IEEE code is deliberately brief, listing ten concise commitments.
- Algorithmic bias: The ACM code explicitly addresses fairness in algorithmic systems and data-driven decision-making (principle 1.4, 2.5), which is a more modern concern. The IEEE code addresses discrimination in more general human-relations terms.
- Privacy: The ACM code dedicates an entire principle (1.6) to privacy as a fundamental value tied to autonomy. The IEEE code treats it more implicitly under public welfare.

2. Case Description

The Predictive Policing Algorithm - A Fictional Case

The following case is fictional but inspired by real-world deployments of algorithmic decision-making in public safety contexts.

UrbanSafe Solutions is a technology startup contracted by the municipal government of a mid-sized European city (population: approximately 500,000) to develop a predictive policing system called PATROL-AI. The goal is to optimize the deployment of police patrols by predicting which geographic zones are most likely to experience criminal incidents on a given day and time.

The system is developed by a team of four software engineers, led by a senior engineer named Marta. The algorithm ingests several years of historical crime reports, socioeconomic census data, real-time CCTV feeds, and weather data. It uses a machine learning model trained on these inputs to generate daily “heat maps” ranking city zones by predicted crime probability. Patrol units are then dispatched according to these rankings.

After the first full year of deployment, the city’s safety committee presents impressive results: overall reported crime rates have fallen by 18%, emergency response times have improved by 12%, and the police department estimates it has saved approximately 900,000 euros in resource allocation. The project is celebrated publicly, and the contract is renewed.

However, during routine model auditing six months into deployment, Marta discovers a serious structural flaw. Three neighborhoods - home to approximately 15,000 residents, predominantly of lower socioeconomic status and with large immigrant and ethnic minority populations - are consistently flagged as high-risk by the algorithm at a rate disproportionate to actual crime rates. The model has learned to replicate historical over-policing patterns present in the training data.

The consequences are measurable and documented internally:

- Residents in these neighborhoods are subjected to police stops and identity checks at a rate 3.4 times higher than similar incidents would warrant based on actual crime statistics.
- Young men in these areas report a significant deterioration in their relationship with authorities and describe a loss of freedom of movement.

- A feedback loop has been identified: increased patrols produce more arrests and incident records in the flagged zones, which further reinforces the algorithm's prediction that those zones are dangerous, regardless of actual underlying crime trends.
- The CCTV data was collected from public cameras operated under a municipal surveillance agreement that predates GDPR and does not explicitly authorize its use in automated profiling systems.

Marta formally documents these findings in an internal technical report and presents it to her direct manager and the company's Chief Technology Officer. Both acknowledge the findings but argue that the aggregate crime reduction benefit justifies the current approach. They argue: "The numbers speak for themselves - the city is safer overall. A few complaints from one part of town don't outweigh that." Marta is instructed to continue development of the next model version without modifying the current deployment.

Marta now faces a decision: comply and stay silent, attempt quiet internal remediation, or escalate externally.

3. Ethical Analysis

3.1 Analysis Under the ACM Code of Ethics

Principle 1.1 (Contribute to society and human well-being): PATROL-AI contributes positively to the well-being of the majority of the city's population through measurable crime reduction. However, it demonstrably reduces the well-being of the 15,000 residents in the over-policed neighborhoods. The ACM code specifies that "all people are stakeholders," which means the interests of all community members - including this minority - must be weighed, not discarded because they are outnumbered.

Principle 1.2 (Avoid harm): The internal audit clearly documents harm to the affected residents: loss of freedom of movement, harassment by authorities, psychological distress, and reputational damage to their communities. The management's instruction to continue deployment despite documented harm is a direct violation of this principle. The ACM code states that if an action produces unavoidable harm, it should be minimized and those harmed should be informed. Neither condition is met here.

Principle 1.4 (Be fair and non-discriminatory): This principle is the most directly violated. The algorithm's differential treatment of residents based on historically biased data - which correlates strongly with race, ethnicity, and socioeconomic status - constitutes algorithmic discrimination. The ACM code explicitly notes that technologies must not be used to discriminate unfairly, whether intentionally or through negligent design.

Principle 1.6 (Respect privacy): The use of CCTV data from a pre-GDPR surveillance agreement in automated profiling likely exceeds the original consent framework. This represents a repurposing of personal data without informed consent, violating the privacy principle.

Principle 2.5 (Comprehensive evaluation): Marta's audit demonstrates professional diligence in line with this principle. However, her managers' dismissal of a documented technical finding represents a failure at the organizational level. A proper evaluation was completed; it was simply ignored.

3.2 Analysis Under the IEEE Code of Ethics

Clause 1 (Public safety and welfare paramount): The IEEE code is unambiguous that the welfare of the public must come first - above corporate interests, contractual obligations, or aggregate statistics. "The public" includes the residents of the over-policed neighborhoods. Management's argument that 18% crime reduction justifies ongoing harm to a minority is in direct conflict with this clause.

Clause 7 (Honest criticism and correction of errors): Marta's internal report is an act of professional integrity consistent with this clause. Her management's failure to accept the criticism and act on the technical findings violates Clause 7. The IEEE code requires that errors be acknowledged and corrected; instead, the error is being maintained by deliberate inaction.

Clause 8 (Fair treatment regardless of origin): The differential impact of the algorithm on ethnic minority and immigrant communities constitutes a failure of fair treatment. Even without discriminatory intent, IEEE ethics requires engineers to be aware of and act to prevent discriminatory outcomes of their technical systems.

4. Utilitarian Analysis, Opinions, and Conclusions

4.1 Applying the Utilitarian Framework

Utilitarianism, as articulated by Jeremy Bentham and refined by John Stuart Mill, holds that the morally correct action is the one that maximizes the total net happiness (or well-being) across all affected parties. To evaluate the case of PATROL-AI, we must perform a utilitarian calculus.

Benefits of the current deployment (in favour):

- Approximately 485,000 residents benefit from a measurable 18% reduction in crime - greater personal safety, reduced property loss, and reduced fear.
- Improved police response efficiency creates savings that could be reinvested in other public services.
- The city's management and company see reputational and financial gains.

Harms of the current deployment (against):

- Approximately 15,000 residents are subjected to systematic over-surveillance and harassment with a documented 3.4x disproportionate stop rate.
- The feedback loop means the harm is not static but growing: over time, the affected neighborhoods accumulate more algorithmic "evidence" of dangerousness, increasing rather than decreasing the injustice.

- GDPR non-compliance exposes all 500,000 citizens to legal risk and erodes the trust citizens should have in how their data is used by government.
- Institutional distrust undermines long-term public safety: communities that distrust police cooperate less with investigations, reducing the very outcomes the system is meant to improve.

4.2 The Fundamental Problem with Management's Argument

Management's position - that the 18% aggregate crime reduction justifies the harm to the minority - is a simplistic, "brute" utilitarian argument. It treats all affected parties as a single pool and concludes that the majority wins. However, even within utilitarianism, this reasoning is flawed for three reasons:

- **Long-term calculus:** The feedback loop means the harm will compound. A genuine utilitarian analysis must account for future harm, not just present benefit. When modelled over 5 years, the concentrated harm to the 15,000 affected residents is likely to exceed the marginal crime-reduction benefit to the majority.
- **Rule utilitarianism (Mill):** Mill argued that we should follow rules which, if universally adopted, produce the best outcomes. The rule "disregard the civil liberties of a minority if it benefits the majority" is a catastrophically dangerous norm. Societies that adopt this rule - even in technical systems - tend toward systematic injustice. A utilitarian committed to good rules would reject it.
- **Intensity of harm:** Bentham's hedonic calculus includes the intensity of pleasure and pain. The harm to those 15,000 residents - loss of freedom, constant suspicion, psychological harm - is severe and concentrated. The benefit to each of the 485,000 others is marginal and diffuse. Intensity-weighted utilitarianism does not clearly favour the current deployment.

4.3 What Marta Should Do - A Utilitarian Recommendation

From a utilitarian standpoint, Marta has a clear professional and moral obligation. The action that maximizes total well-being is not silence and not resignation, but active remediation:

- Immediately document all findings with timestamps and escalate formally, creating a record that protects both herself and the public interest.
- Propose a specific technical remediation plan: bias correction in the training dataset, introduction of fairness constraints in the model, and independent auditing before the next deployment cycle.
- If internal escalation is exhausted without result, Marta should consider contacting the relevant supervisory authority for GDPR compliance (the national data protection authority) - not as an act of disloyalty, but as the action most likely to produce the best outcome for the most people.

The utilitarian case for speaking up is strong: the short-term harm to UrbanSafe Solutions (reputational, financial) is significantly smaller than the long-term harm to 15,000 residents and to public trust in algorithmic governance if the issue goes unaddressed. Greater good requires Marta to act.

4.4 Team Conclusions

This case illustrates a tension that is increasingly common in the technology industry: aggregate statistical benefit is used to justify concentrated structural harm. Both the ACM and IEEE codes, as well as a careful utilitarian analysis, converge on the same conclusion - the current deployment of PATROL-AI is ethically unacceptable not because it has no benefits, but because it achieves those benefits through means that violate the dignity, rights, and welfare of a specific community.

The engineers involved - and Marta in particular - are not absolved of responsibility by following management's instructions. Both codes explicitly place individual professional responsibility above organizational authority when public welfare is at stake. The IEEE code states that engineers must hold public safety "paramount," and the ACM code requires professionals to escalate when reasonable internal efforts to resolve ethical concerns fail.

In our own assessment, the most important lesson of this case is that utilitarian reasoning is easily weaponized when only aggregate numbers are examined and the distribution of harm is ignored. Ethical analysis - whether guided by professional codes or moral philosophy - must always ask not just "how much good is produced?" but also "who bears the cost, and is that distribution just?"

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